THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A concave mirror held close to the face shows an image that is: 1

(a) real and inverted **(b)** real and diminished
(c) virtual, erect and enlarged **(d)** virtual and diminished

2. After about 3 years, the lunar calendar falls behind the solar by about: 1

(a) 33 days **(b)** 1 day **(c)** 22 days **(d)** 11 days

3. Which is the main cause of the rise in the CO2 level in the atmosphere? 1

(a) pollination
(b) photosynthesis
(c) rainfall
(d) burning of fossil fuels and deforestation

4. Mars is cold because it: 1

(a) has oceans
(b) has a thick atmosphere
(c) is farther from the Sun and has a thin atmosphere
(d) is nearer to the Sun

5. To test whether the type of soil affects plant growth, which factor should be varied? 1

(a) type of soil **(b)** seed type **(c)** amount of light **(d)** amount of water

6. Which two are found in both a plant and an animal cell? 1

(a) nucleus and cytoplasm **(b)** chloroplast and wall
(c) wall and vacuole **(d)** chloroplast and nucleus only

7. Mucus in the nose helps by: 1

(a) producing vitamins **(b)** making noise
(c) digesting food **(d)** trapping dust and germs

8. A wire carrying current produces a field in the form of: 1

(a) no pattern **(b)** circles around the wire
(c) dots **(d)** straight lines along the wire

9. A magnet is cut into two pieces. Each piece will: 1

(a) be a south pole only **(b)** lose magnetism
(c) have two poles **(d)** be a north pole only

10. The speed of a cyclone's wind is highest in the: 1

(a) eye **(b)** the centre **(c)** eye wall **(d)** outer edge only

11. Why does a closed container of water not appear to lose water? 1

(a) pressure is low
(b) water is solid
(c) no evaporation occurs
(d) vapour cannot escape; condensation balances evaporation

12. The cooking utensils are made of metals because they are: 1

(a) poor conductors **(b)** brittle
(c) lighter than water **(d)** good conductors of heat

13. The solubility of salt at 20 degrees Celsius is 36 g per 100 g of water. If 40 g is added to 100 g of water at 20 degrees Celsius, then: 1

(a) all dissolves **(b)** 40 g remains
(c) 4 g remains undissolved **(d)** 36 g remains

14. An object is placed very far from a convex lens. The image forms: 1

(a) at the lens **(b)** far on the same side
(c) at infinity **(d)** at the focus

15. The new Moon rises at about: 1

(a) sunrise **(b)** sunset **(c)** noon **(d)** midnight

16. The disappearance of vultures leads to a rise in: 1

(a) tigers **(b)** stray dogs and diseases
(c) plants **(d)** cows

17. Scientists get information about the interior of the Earth from: 1

(a) photographs of the core **(b)** digging to the core
(c) telescopes **(d)** seismic waves of earthquakes

18. If 20 swings take 36 s, the time period of one swing is: 1

(a) 720 s **(b)** 18 s **(c)** 1.8 s **(d)** 0.18 s

19. Which of these statements about bacteria is correct? 1

(a) They are visible to the naked eye.
(b) They have a nucleus and chloroplasts.
(c) They have genetic material but no true nucleus.
(d) They are always harmful.

20. A student recovered from chickenpox and does not get it again. This is due to: 1

(a) immunity **(b)** exercise **(c)** fever **(d)** diet

21. An electromagnet lifts 10 clips with 20 turns. With 40 turns of the same wire and cell, it will lift: 1

(a) no clips **(b)** the same number of clips
(c) more clips **(d)** fewer clips

22. A ball on a table has weight 5 N. The table pushes it with: 1

(a) 5 N upward **(b)** 5 N downward **(c)** 10 N **(d)** 0 N

23. A person of weight 600 N stands on both feet, each of area 0.01 square metre. The pressure on the ground is: 1

(a) 60000 Pa **(b)** 300 Pa **(c)** 6000 Pa **(d)** 30000 Pa

24. The smell of incense sticks reaches the farthest room slowly because: 1

(a) diffusion in air takes time **(b)** incense does not diffuse
(c) air is a solid **(d)** incense is a liquid

25. The ratio of hydrogen to oxygen by mass in water is: 1

(a) 2 : 1 **(b)** 1 : 2 **(c)** 1 : 8 **(d)** 8 : 1

26. A solution contains 15 g of solute in 135 g of solvent. The mass percentage is: 1

(a) 15 per cent **(b)** 10 per cent **(c)** 9 per cent **(d)** 11 per cent

27. A girl stands 1.5 m from a plane mirror. If she walks 0.5 m towards it, the distance between her and the image is: 1

(a) 1 m **(b)** 2 m **(c)** 3 m **(d)** 0.5 m

28. If the new moon is on 1 January, the next full moon will be on about: 1

(a) 15 January **(b)** 8 January **(c)** 30 January **(d)** 1 February

29. In the food web, an organism that eats both plants and animals occupies: 1

(a) no level **(b)** only the third
(c) only the first **(d)** more than one trophic level

30. At higher altitudes, breathing is difficult because: 1

(a) oxygen is poisonous
(b) air is thinner with lower oxygen amount per breath
(c) air is thicker
(d) air is warm

31. An observation that cannot be repeated by others is: 1

(a) unreliable **(b)** a law **(c)** proven **(d)** reliable

32. A scientist sees under a microscope a tiny moving single-celled organism with a nucleus in pond water. It is likely to be a: 1

(a) bacterium **(b)** protozoan **(c)** virus **(d)** mushroom

33. Which is a lifestyle disease? 1

(a) measles **(b)** cholera **(c)** type 2 diabetes **(d)** chickenpox

34. A thick copper wire is chosen for home wiring because it: 1

(a) heats less due to low resistance **(b)** is a magnet
(c) melts easily **(d)** is an insulator

35. If a negatively charged rod is brought near a neutral metal disc, the leaves: 1

(a) open **(b)** stay closed
(c) become charged permanently **(d)** burn

36. In a low pressure region air is: 1

(a) frozen **(b)** rising **(c)** stationary **(d)** sinking

37. A student sprinkles water on a hot surface and it disappears. The process is: 1

(a) melting **(b)** freezing **(c)** evaporation **(d)** condensation

38. Which element conducts electricity but is a non-metal? 1

(a) bromine **(b)** graphite (carbon)
(c) sulphur **(d)** phosphorus

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39. A mixture that is cloudy and settles on standing is: 1

(a) a colloid **(b)** a suspension **(c)** a true solution **(d)** an element

40. Rays parallel to the principal axis, after reflection from a concave mirror, pass through the: 1

(a) pole **(b)** principal focus

(c) edge of the mirror **(d)** centre of the object

41. The Hindu lunisolar calendar adds an extra month called: 1

(a) Adhik maas **(b)** Magha **(c)** Pausha **(d)** Shravana

42. Plants lose water vapour through tiny pores called: 1

(a) petals **(b)** stomata **(c)** roots **(d)** nodules

43. Which of these would you NOT regard as a renewable source of water? 1

(a) rivers **(b)** fossil groundwater in deep layers

(c) lakes **(d)** rainfall

44. Which question needs measurement to be answered? 1

(a) How long does ice take to melt in a warm room?

(b) Do you like ice?

(c) Is ice cold feeling good?

(d) Is ice pretty?

45. What is the function of the nucleoid in bacteria? 1

(a) makes food **(b)** provides shape

(c) stores water **(d)** contains genetic material

46. Misuse of antibiotics leads to: 1

(a) resistant bacteria **(b)** stronger bones

(c) quicker recovery from viral infections **(d)** better vaccines

47. An electric fuse of a house is rated 15 A. Appliances of 2 A, 5 A, 6 A and 4 A are connected together. The fuse will: 1

(a) become hot only **(b)** melt as 17 A exceeds 15 A

(c) stay intact **(d)** become a magnet

48. A small iron nail is attracted by a magnet at 2 cm distance. This shows the force is: 1

(a) a contact force **(b)** a non-contact force

(c) muscular **(d)** friction

49. The eye of the cyclone is calm because: 1

(a) winds are strong **(b)** air rises

(c) rain is heavy **(d)** air sinks at the centre

50. Water in a closed pot of a pressure cooker boils above 100 degrees Celsius because: 1

(a) increased pressure raises the boiling point

(b) pressure is lower

(c) heat is lost

(d) water is heavier

END OF QUESTION PAPER **Total: 50 Marks**